Topics 7.11 – 7.12: MCQ Practice

1. The value of K_{sp} for the salt BaF_2 is 1.84×10^{-7}. The $[F^-]$ in a saturated solution of BaF_2 is closest to

   (A) $4.29 \times 10^{-4}\ M$

   (B) $3.58 \times 10^{-3}\ M$

   (C) $4.51 \times 10^{-3}\ M$

   (D) $7.17 \times 10^{-3}\ M$

Compound	K_{sp}
$Cd(OH)_2$	7.2×10^{-15}
$Mg(OH)_2$	5.6×10^{-12}
$Ni(OH)_2$	5.5×10^{-16}

2. Which of the following ranks the compounds listed in the table above in order of increasing solubility?

   (A) $Ni(OH)_2 < Cd(OH)_2 < Mg(OH)_2$

   (B) $Ni(OH)_2 < Mg(OH)_2 < Cd(OH)_2$

   (C) $Cd(OH)_2 < Ni(OH)_2 < Mg(OH)_2$

   (D) $Mg(OH)_2 < Cd(OH)_2 < Ni(OH)_2$

Compound	K_{sp} at 298 K
Ag_2SO_4	1×10^{-5}
$PbSO_4$	1×10^{-8}

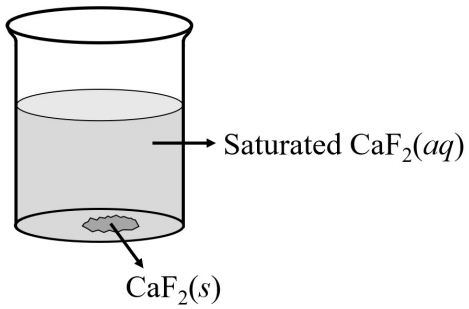
3. A 1.0 L solution of $AgNO_3(aq)$ and $Pb(NO_3)_2(aq)$ has a Ag^+ concentration of 0.020 M and a Pb^{2+} concentration of 0.0010 M. A 0.0010 mol sample of $K_2SO_4(s)$ is added to the solution. Based on the information in the table above, which of the following will occur? (Assume that the volume change of the solution is negligible.)

   (A) No precipitate will form.

   (B) Only $Ag_2SO_4(s)$ will precipitate.

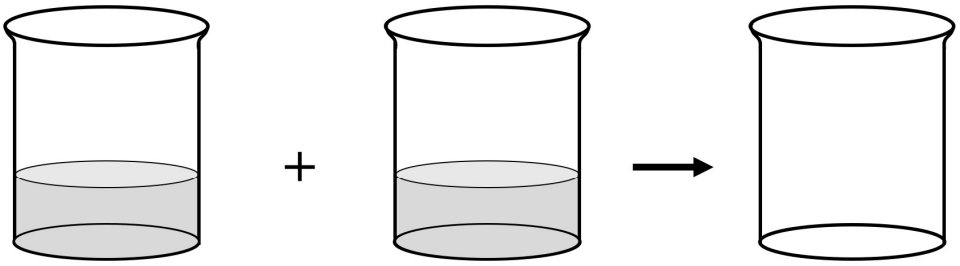
   (C) Only $PbSO_4(s)$ will precipitate.

   (D) Both $Ag_2SO_4(s)$ and $PbSO_4(s)$ will precipitate.



4. The saturated CaF_2 solution shown above is left uncovered on a lab counter at a constant temperature. Evaporation occurs slowly over a period of one week, and the total volume of the solution decreases. Which of the following choices correctly identifies what happens to the value of $[Ca^{2+}]$ in the solution during this time period and provides the correct justification?

	What happens to $[Ca^{2+}]$	Justification
(A)	$[Ca^{2+}]$ increases	As water evaporates, some of the $CaF_2(s)$ on the bottom of the beaker dissolves in the solution.
(B)	$[Ca^{2+}]$ increases	As water evaporates, more $CaF_2(s)$ precipitates out of the solution in the beaker.
(C)	$[Ca^{2+}]$ remains the same	As water evaporates, some of the $CaF_2(s)$ on the bottom of the beaker dissolves in the solution.
(D)	$[Ca^{2+}]$ remains the same	As water evaporates, more $CaF_2(s)$ precipitates out of the solution in the beaker.



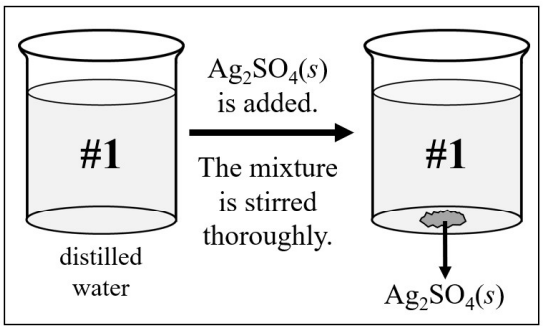
5. A student combines 50.0 mL of 0.040 *M* $Pb(NO_3)_2(aq)$ with 50.0 mL of 0.040 *M* NaCl(*aq*). The mixture is stirred to ensure that it is thoroughly mixed and has a final volume of 100.0 mL. The value of K_{sp} for $PbCl_2$ is 1.7×10^{-5}. Which of the following will occur in this experiment and why?

(A) $PbCl_2(s)$ will precipitate because $Q > K_{sp}$.

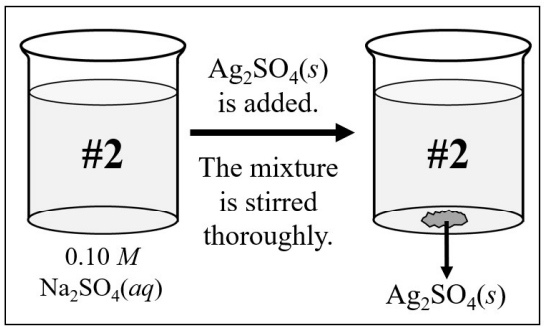
(B) $PbCl_2(s)$ will precipitate because $Q < K_{sp}$.

(C) $PbCl_2(s)$ will not precipitate because $Q > K_{sp}$.

(D) $PbCl_2(s)$ will not precipitate because $Q < K_{sp}$.



$Ag_2SO_4(s)$ is added to a sample of distilled water in beaker #1.



$Ag_2SO_4(s)$ is added to a sample of 0.10 *M* $Na_2SO_4(aq)$ in beaker #2.

6. A student prepared two saturated solutions by adding an excess amount of $Ag_2SO_4(s)$ to two different beakers, as shown in the diagram above. Beaker #1 contained distilled water, and beaker #2 contained 0.10 *M* $Na_2SO_4(aq)$. The contents of each beaker were stirred thoroughly after the addition of $Ag_2SO_4(s)$. At the end of the experiment, a small amount of undissolved $Ag_2SO_4(s)$ was present in each beaker. Which of the following choices correctly compares the value of $[Ag^+]$ in each beaker at the end of the experiment and provides the correct justification?

	Comparison of $[Ag^+]$	Justification
(A)	$[Ag^+]$ in beaker #2 is less than $[Ag^+]$ in beaker #1.	The solubility of Ag_2SO_4 is reduced in beaker #2 because of the presence of a common ion, SO_4^{2-}.
(B)	$[Ag^+]$ in beaker #2 is less than $[Ag^+]$ in beaker #1.	The solubility of Ag_2SO_4 is increased in beaker #2 because of the presence of a common ion, SO_4^{2-}.
(C)	$[Ag^+]$ in beaker #2 is greater than $[Ag^+]$ in beaker #1.	The solubility of Ag_2SO_4 is reduced in beaker #2 because of the presence of a common ion, SO_4^{2-}.
(D)	$[Ag^+]$ in beaker #2 is greater than $[Ag^+]$ in beaker #1.	The solubility of Ag_2SO_4 is increased in beaker #2 because of the presence of a common ion, SO_4^{2-}.

$$BaCO_3(s) \rightleftarrows Ba^{2+}(aq) + CO_3^{2-}(aq)$$

7. The dissolution of the slightly soluble salt $BaCO_3$ is represented by the equation above. Which of the following changes will decrease $[Ba^{2+}]$ in a saturated solution of $BaCO_3$, and why? (Assume that after each change some $BaCO_3(s)$ remains in contact with the solution.)

(A) Allowing some of the water to evaporate from the solution, because more $BaCO_3(s)$ will precipitate

(B) Adding a strong acid, because some $CO_3^{2-}(aq)$ ions will be converted into $CO_2(aq)$ molecules

(C) Adding $BaCO_3(s)$, because the value of Q will become greater than K_{sp}

(D) Adding $K_2CO_3(s)$, because the reaction will proceed toward reactants